Digital Transformation Technologies in Insurance: A Bibliometric Analysis and Systematic Literature Review

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Abstract

The insurance market's employing of digital transformation approaches is a quickly developing subject, but further research is needed for Mapping the Research Landscape to fully understand the long-term effects and real repercussions on the insurance value chain. This work uses an innovative strategy that combines bibliometric analysis with a systematic literature review (SLR) that adheres to the PRISMA framework to close this gap. In the first step, which involved a thorough bibliometric study, important countries were selected, and publications, journals, keywords, affiliations, and significant researchers in the field of DT in insurance contexts were distributed. A bibliometric analysis help track changes over time and provide a longitudinal perspective on how the adoption and impact of digital technologies have progressed in insurance. This phase offered a broad overview of the field, highlighting the global and interdisciplinary dimensions of DT research within the insurance industry. The subsequent phase involved a meticulous SLR of 37 select scholarly articles. This in-depth review sheds light on the impacts of digital transformation technologies insurance, Trend Identification, important variables, DT technologies have the most used, Gaps and future directions of DT in insurance. Underscoring the revolutionary potential of DT in the insurance industry, the findings from this dual-method approach provide a thorough roadmap for insurers, academics, and legislators. The review's vast collection of articles offers a thorough examination of the varied and substantial effects of DT in the insurance industry, stressing its function in areas like estimating insurance claims, premiums, health insurance fraud, driving behavior analysis, risk assessment, and preparing the next generation for DT's integration in a variety of industries like asset and machinery insurance.

Keywords

Digital Transformation; Machine learning; Digital twins; IoT; Predictive Maintenance; Insurance Risk Management; http://www.ijimai.org/journal/biblio/keywords

I. Introduction

INSURANCE providers are middlemen that aggregate several similar risks and oversee the payment process within that risk pool to offer protection and financial coverage against unforeseen disasters [1]. Per the guidelines for internal audit, internal control, and risk management in insurance companies (Circular: 70/2022/TT-BTC). Identifying, quantifying, tracking, and controlling risks is the responsibility of risk management in the market for insurance. An essential component of the insurance sector is risk assessment [2]. Regulations require insurers to accurately and promptly identify, assess, monitor, and control risks [2]. Nonetheless, it is often static and carried out at the start of a client contract. Even though continuous risk factor assessment is well known, it is not frequently applied to insurance product personalization [3]. Property insurance risk management is a crucial aspect of protecting assets and mitigating potential financial losses for individuals and businesses.

Property insurance provides coverage for various types of properties, including machines, homes, commercial buildings, industrial facilities. However, the risks associated with property ownership and potential damage can be significant, ranging from natural disasters such as fires, floods, and storms to incidents like breakdown or failure, theft, vandalism, and accidents [4]-[5]. Detecting, evaluating, and reducing any risks to properties, as well as putting protective measures in place, are all necessary for effective property insurance risk management. Following the identification and evaluation of the risks.

Risk management plans are created and put into action. These strategies may include [6]:

- *Risk Prevention and Mitigation*: This may involve installing security systems, fire sup-pression systems …etc.
- *Loss Control Measures*: This may include regular maintenance, inspections …etc.
- *Ongoing Monitoring and Review*: Risk management is a continual process that must be continuously monitored and reviewed to make certain that it's still current and effective.

In recent years, we have seen an exponential growth in digital transformation technology such as Machine Learning, Expert System, Big Data, Digital Twins and Internet of Things (IoT). In the context of the insurance industry [7]-[8]-[9].

Most insurance sector operations, such as risk management, the insurance value chain, product development and insurance policies, underwriting and claims, distribution and sales, fees, and asset liability management, are impacted by digital innovation [10]-[11]. Insurers must increase their technological hardware and software to support value chain transformation, process optimization, and operational efficiency to increase their competitiveness [12].

Insurtech, or digital transformation (DT), is enhancing the insurance industry and promoting significant growth in underwriting and claims verification, marketing, product development, risk management, and fund utilization [13]-[14]. The technological revolution in insurance services is not only inevitable but also serves as a key driver of progress and innovation. Technological developments in the insurance sector have made it possible to create new business models and highlight the sector's digital transformation [15]. These developments also result in the exclusion of businesses that cannot keep up with the changes. Therefore, to take advantage of the prospects for development and problems presented by the new digital technologies, businesses must innovate and reconsider their business models [16]. They also need to change their processes and products.

II. Research Problem

While insurance provides valuable financial coverage as a method for risk financing, it may not be suitable for all types of risks faced by businesses and industries. Insurance policies often cover property damage and may provide some coverage for business interruption losses. However, the financial impact of a business interruption can extend beyond the direct costs of property damage. Additional losses, such as lost profits, increased expenses, and supply chain disruptions, may not be fully covered by insurance. Some risks require a proactive approach that goes beyond financial compensation, necessitating the development of alternative strategies.

Insurance organizations are facing a significant challenge in the digital age due to the unprecedented number of competitive challenges posed by technical developments. One concern is that certain areas may see a decline in the demand for insurance. Volvo and Mercedes, for example, are so sure of their self-driving vehicles that they are not purchasing collision coverage [4]. Insurance businesses must expedite the digitization of their operational and commercial operations. Conventional insurance firms still have a poor understanding of digital technologies as they primarily rely on sales channels, brokers, and physical intermediaries [17]. The adoption of digitalization by insurers was slow [18].

Where few companies bet on the creation of value and incorporation of new technologies. While there is academic and industry interest in digitalization and emerging technologies in insurance, the impact of DT on the Property and Casualty (P&C) insurance sector has been understudied so far [4]. This includes exploring how technologies like artificial intelligence, machine learning, digital twins, and Internet of Things (IoT), can be leveraged to enhance various aspects of the risk management of property insurance.

The paper offers valuable insights for the insurance industry and researchers by reinforcing gaps and future directions, as well as summarizing existing evidence on the role of digital transformation within the insurance sector. Distinguished by its comprehensiveness, this paper provides a broad and forward-looking perspective through bibliometric analysis and systematic review. Thus, it represents a new development in the discipline of insurance technology research.

III. Research Questions

To identify appropriate keywords and specific search strings for addressing research questions, the study employed the PICO criteria, which stands for Population, Intervention, Comparison, Outcomes, in addition to Language, and Timing.

Table 1 PICO criteria

Population	Insurance companies.
Intervention	Digital Transformation Technologies processes at insurance companies.
Comparison	In this investigation, no comparative intervention is anticipated.
Outcomes	Enhanced precision in risk evaluation through Digital Transformation technology.
Timing	Over a 5-year period.
Language	Articles were reported in English as it is a worldwide language, preventing prejudice.

This review is split into two main sections—firstly, a systematic review of the literature, and secondly, bibliometric analysis to achieve the study's objectives. These enquiries relate to bibliometric analysis, which is the beginning stage of the review.

1) What is the distribution of studies per year, leading countries, prominent journals, key authors, and most common keywords related to the integration of digital transformation (DT) in the insurance sector?

The following questions are specific to the second stage, which is systematic review:

2) What are the influences of DT technologies in the insurance market?
3) Which digital transformation technology has the most used on different segments of the insurance industry?
4) What are the key factors influencing the insurance industry's digital transformation?
5) What gaps in the literature exist, and what future research directions for digital transformation in insurance?

IV. Research Structure

The review paper is structured to provide a thorough analysis of the topic, beginning with an introduction that establishes the relevance and objectives of the review. it then clearly articulates the research problem, defining the issue at hand, explaining its significance, and situating it within the existing literature. Following this, the research questions are outlined, detailing both primary and secondary questions that guide the investigation and offering a focused framework for exploration.

The theoretical background section presents a comprehensive overview of pertinent theories and conceptual frameworks, situating the review within a broader academic context. The surveyonally, the meta-survey summarizes trends from previous reviews, while the review methodology discusses the search strategy, inclusion and exclusion criteria, and methods for data extraction and quality assessment.

The results section synthesizes findings, highlighting key insights and emerging patterns. finally, the conclusion summarizes the main findings, discusses their implications, acknowledges limitations, and suggests directions for future research. This organized approach ensures a coherent and detailed examination of the topic, as illustrated in Fig 1.

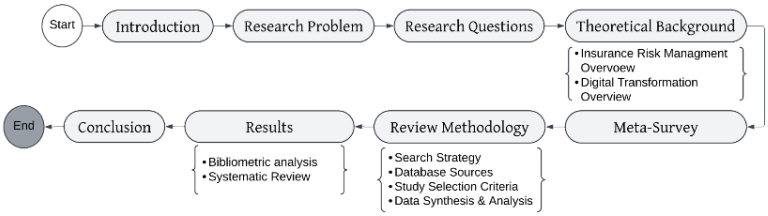

Fig. 1 Structure and organization of the review paper

V. Theoretical Background

A. *Insurance* Risk Management *overview*

This section provides an overview of the definitions of insurance and risk management, along with the various types associated with each.

1. *Insurance definition & Types of Insurance*

Insurance is a contractual arrangement between a policyholder and an insurer, where the policyholder pays a premium in exchange for financial protection against potential losses or risks. It includes the policyholder's shift of risk to the insurer, who agrees to provide compensation or benefits in the event of covered losses. Insurance can be broadly categorized into several types; each designed to address specific risks and provide financial protection in different areas of life [19]-[20]. as classified in Fig 2.

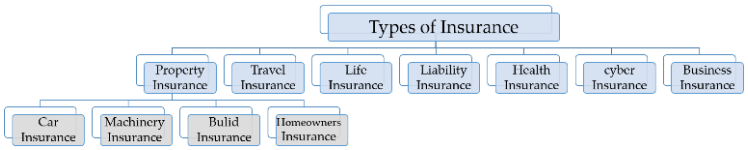

Fig. 2 Insurance products overview

- *Life Insurance*: offers financial protection to the beneficiaries named by the policyholder upon their death. It provides a death benefit that can assist with funeral costs, replace lost income, settle outstanding debts, or support dependents financially [21].
- *Health Insurance*: reimburses medical expenses incurred by the policyholder or their covered dependents. Prescription drugs, hospital stays, doctor visits, and other healthcare services are all partially covered by it. One can receive health insurance through private plans, government programs, or through one's work [22].
- *Auto Insurance*: vehicle insurance protects against loss or damage caused by theft, collisions, and other events. It usually consists of three types of coverage: comprehensive (for loss or damage not due to a collision), collision (for damage to the insured vehicle), and liability (for third-party injuries and property damage) [23].
- *Homeowners Insurance*: homeowners insurance guards against loss or damage from calamities like fire, theft, vandalism, or natural catastrophes to the policyholder's house and possessions. It usually includes liability for injuries sustained on the property, personal items, the home's construction, and supplementary living costs in the event that the house becomes unfit [24].
- *Business Insurance*: business insurance provides coverage for various risks faced by businesses. It includes property insurance (for buildings, equipment, and inventory), liability insurance (for injuries or damages caused by the business), business interruption insurance (for income loss due to a covered event), and specialized coverage for specific industries or professions [25].
- Cyber insurance is a type of insurance product aimed at safeguarding businesses against cyber incidents. Its primary purpose is to cover or mitigate the financial losses incurred by the insured because of such incidents [26].
- *Machinery insurance*: which is also referred to as equipment or machinery insurance, offers coverage for diverse machinery and equipment utilized in business activities. It serves as protection against unexpected occurrences like accidents, mechanical malfunctions, natural calamities, vandalism, or theft, safeguarding insured machines from damage, breakdowns, or loss [27].
- *Travel Insurance* provides protection against unforeseen circumstances that may arise while travelling domestically or abroad. It can include trip cancellation/interruption coverage, emergency medical expenses, lost luggage coverage, and other benefits to mitigate financial risks while traveling [25].

Home, car, machinery, and building insurance all fall under the broader umbrella of property insurance. Each of these types of insurance is designed to protect specific as-sets against various risks and perils. These diverse types of insurance allow individuals, businesses, and organizations to manage risks, protect assets, and gain peace of mind in the face of uncertainties.

2. *Risk Management overview*

Insurance Risk Management is a discipline within the insurance industry that focuses on identifying, assessing, and managing risks associated with insurance activities. It involves the implementation of strategies and measures to mitigate the potential financial losses that may arise from these risks.

Risk is generally defined as the uncertainty of outcomes, often associated with negative events or losses. In the context of insurance, risk refers to the possibility of an event occurring that could result in financial loss for the insurer or policyholders. Insurance companies assume various risks in their operations, and effective risk management is crucial to ensure their long-term sustainability [28].

Risk can be approached through two primary strategies: risk control and risk transfer [5]-[20]-[29]. as classified in Fig 3**.** Risk control encompasses strategies and measures intended at reducing both the frequency and severity of risks. It involves the implementation of various techniques, including avoidance, loss prevention, and loss reduction.

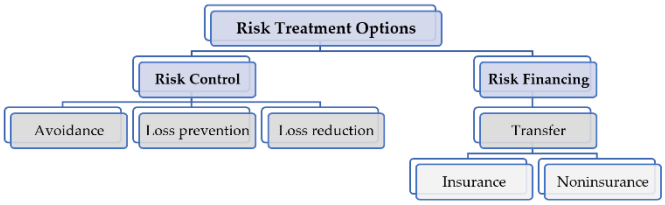

Fig. 3 overview of addressing risk strategies.

- *Loss Reduction*: involves methods and actions put in place to minimize potential losses' extent or effects. This includes using controls, protections, or mitigation methods to decrease the chances or seriousness of losses. For instance, in property insurance, loss reduction can mean installing fire suppression systems, using security measures, or regularly maintaining the property to lower the risk of damage and financial loss [30].
- *Loss Prevention:* includes taking proactive steps to minimize the likelihood or impact of potential losses. This could contain actions like installing security systems, implementing safety measures, or conducting regular maintenance to reduce the risk of property damage [31].

- *Avoidance:* refers to the practice of completely sidestepping insuring high-risk properties or declining coverage for activities with significant potential for losses. Insurance companies may opt to avoid certain risks altogether if they deem them too risky or costly to manage effectively, thus avoiding the potential losses associated with those risks [30].

Risk transfer involves the shifting of the financial consequences of specified losses to another party. This can be achieved through insurance and non-insurance techniques.

- Insurance serves as a common form of risk transfer, where policyholders transfer the financial risk to insurance companies in exchange for premium payments.
- Non-insurance techniques, such as contracts, indemnity agreements, or outsourcing, are also used to transfer the financial consequences of losses to another party.

B. Digital transformation overview

This section offers an overview of the concept of digital transformation and the essential technologies that play a role in it.

1. Digital transformation concept

Adopting and utilizing digital technology to radically improve many aspects of an organization's operations, business procedures, and customer interactions is known as digital transformation. A major change in the way the organization functions and generates value is the outcome of this process, which involves integrating digital technology across the board [32]- [33].

Typically, digital transformation involves adopting and employing technologies like cloud computing, artificial intelligence, big data analytics, the Internet of Things (IoT), mobile applications, and automation [34]. The primary objective is to improve efficiency, flexibility, innovation, and customer focus, thereby stimulating growth and gaining a competitive edge in the digital era. Digital transformation encompasses not only technological shifts but also cultural and organizational changes [17]. This aims to embrace digital capabilities and instill a digital-first mindset throughout the organization.

2. Digital transformation technologies

Various digital transformation technologies find application within the insurance sector. Fig 4 provides an outline of these technologies' landscape, although it does not encompass the entirety of them. The following elaboration highlights the crucial technologies pertinent to the current review [35]- [36].

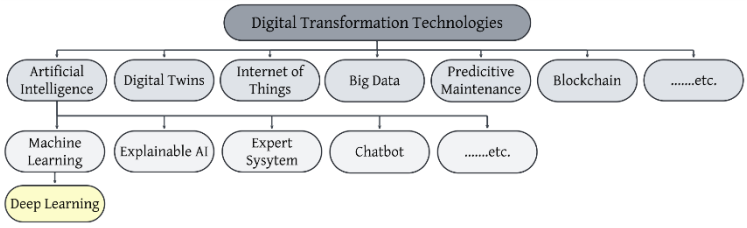

Figure 4 overview of digital transformation technologies

Digital Twins: A digital twin is a dynamic virtual representation of a physical object or system, usually across multiple stages of its lifecycle. It uses real-world data, machine learning models, combined with data analysis, to enable understanding, learning, reasoning, and predicting [37]- [38]. The kind and direction of data flow between the physical and virtual systems is the primary distinction among digital twins, digital shadows, and digital models.

A digital model is an electronic replica of a physical item that is either real or planned; data is not automatically transferred between the digital and physical models. Data is moving from the actual item to the digital model in the digital shadow scenario. A DT has completely integrated data flows in both ways between the physical and digital objects that are now in place [39]-[40].

Artificial intelligence (AI): can be defined as intelligence displayed by robots that replicate cognitive processes related to the human mind, such as thinking, learning, and problem solving. One significant area of AI is machine learning. Through an iterative training procedure, algorithms employ inputs to learn the transformation rules needed to create a desired output by minimizing the error between the generated and the desired result [41]-[9].

Predictive maintenance (PM): is a proactive maintenance strategy that uses data analysis and advanced technologies to predict when equipment or machinery is likely to experience a failure or malfunction. It goes beyond traditional preventive maintenance, which relies on fixed schedules or predetermined intervals for maintenance tasks, regardless of the actual condition of the equipment. Predictive maintenance utilizes real-time data, sensor readings, and analytics techniques to monitor the performance and health of equipment continuously. By analyzing historical data and patterns, predictive maintenance algorithms can identify early warning signs of potential failures, allowing maintenance activities to be scheduled at the optimal time before a breakdown occurs [42]-[43]-[44].

Industrial Internet of Things (IIoT): represents a network of smart and extensively linked industrial elements implemented to attain heightened production rates while minimizing operational expenses. This is achieved through real-time surveil-lance, effective administration, and regulation of industrial procedures, assets, and operational durations [45].

VI. Meta Survey

This section presents several previous reviews related to risk management of insurance using DT technologies in insurance. as shown in Table 2.

In This article by, *Gürsev* analyzed the artificial intelligence products utilized by contemporary insurance businesses and concentrated on the same objective when discussing the deployment of AI techniques in the insurance industry [46].

In a thorough review of the literature on digital technologies within the insurance sector, *Srivastava* highlights both the potential and anticipated changes, examining the benefits, facilitators, and challenges associated with these technologies, especially in the context of advancements driven by Industry 4.0 [47].

The objective of the review paper by *Emer Owens* is to explore the current state of research and practical implementations of XAI techniques in value chain of insurance (Marketing, Product Development, Sales and Distribution, Underwriting and Pricing, Contract Administration and Customer Services, Claim Management, Asset and Risk Management), highlighting their benefits, challenges, and prospects [48].

In this study, *Van* examined the impact and maturity of digital transformation (DT) in the property and casualty (P&C) insurance sector through a mixed-method approach, which included analysing literature, industry reports, and an online questionnaire. The questionnaire was administered to 13 P&C executives from Belgium, South Africa, and the United States. The study found that DT affects every core activity in the P&C insurance value chain, with varying levels of maturity [4].

Table 2 Comparison of other related studies in the same domain: (✓: Yes, ✗: No)

Authors	year	Objective of study	Methodology		Property type		Digital transformation technologies				
			Bibliometric	Review	Vehicle	Equipment	Machine Learning	IOT/ IIOT	Digital Twin	Predictive Maintenance	Data Analytics
Eckert,[41]	2020	Comprehensive overview of digital technologies and the insurance sector's applications that arise from them.	✗	✓	✓	✗	✓	✓	✗	✗	✓
Gürsev,[45]	2023	The study looked at artificial intelligence solutions that are currently in use by insurance firms.	✗	✓	✗	✗	✓	✗	✗	✗	✓
Van,[4]	2021	Examined how DT has affected and matured in the P&C insurance market through a review of relevant literature, industry reports, and an online survey.	✗	✓	✗	✗	✓	✓	✗	✗	✓
Eling,[49]	2018	Apply Porter's value chain framework to analyze the impact of the digital revolution on the insurance industry.	✗	✓	✗	✗	✓	✓	✗	✗	✓
Srivastava, [47]	2022	Exploring the advantages, facilitators, and barriers of digital technologies in the insurance domain.	✗	✓	✗	✗	✓	✓	✗	✗	✓
Owens,[48]	2022	Explore the current state of research and practical implementations of XAI techniques.	✗	✓	✓	✗	✓	✗	✗	✗	✗
Our review	2024	Comprehensiveness study for impacting DT technologies on insurance based on two complementary research methods Bibliometric analysis and systematic review used to assess and synthesize academic literature.	✓	✓	✓	✓	✓	✓	✓	✓	✓

This article by *Eckert* provided a comprehensive review of digital technologies (e.g., blockchain, big data, IoT, AI) and their relevant applications within the insurance sector. To achieve this, a review of academic research, business studies, and publications from regulatory agencies was conducted. By identifying the resulting IT needs for insurers, the article revealed numerous interdependencies among digital technologies [41].

Martin *Eling* led a study aimed at examining the impact of digital transformation on the insurance industry. In this research, Eling and his co-author applied Porter's value chain framework and Berliner's insurability criteria to analyze how digitalization influenced key areas, including product development, underwriting, claims processing, and risk assessment. Their findings were grounded in an extensive analysis of a dataset comprising 84 academic papers and industry reports, offering a comprehensive look at the shifts occurring in the insurance sector due to technological advancements [49].

The current review makes a significant contribution by focusing on digital transformation techniques. specifically, digital twin, Machine learning, Internet of things and predictive maintenance, data analytics, and their application in insurance, while previous reviews primarily emphasized artificial intelligence technologies, the Internet of Things, and data analytics, the current review expands the scope to explore how digital transformation can enable insurance companies to evolve from traditional risk financing model to new business model centered around risk prediction and management.

This research provides valuable insights for the insurance industry and researchers by highlighting gaps and future trends, as well as summarizing existing evidence on the role of digital transformation within the insurance sector.

VII. Review Methodology

This part provides an overview of the approach used for conducting the review. The guidelines established by Kitchenham and Charters [50] were followed. Furthermore, a recent study indicates that it is analogous to the PRISMA method [51]. The literature review was conducted using a suitable database, and a search strategy was devised to identify pertinent studies. Fig. 5 illustrates the procedures in the below-documented systematic literature review approach.

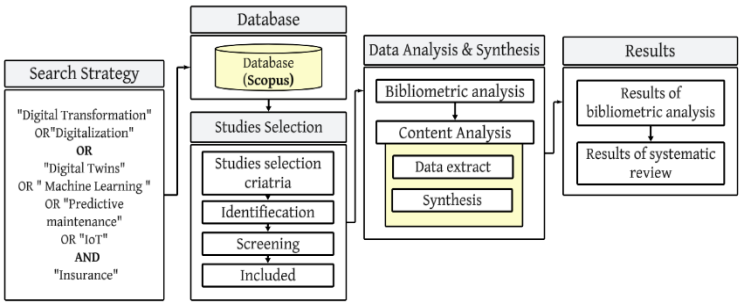

Figure 5 Systematic review methodology: data collection and analysis phases.

A. Search Strategy

Based on Scopus database, A thorough search for peer-reviewed publications on digital transformation in insurance technology was conducted on August 5, 2024. The descriptions of the search parameters we employed are provided below. we utilized the advanced search option in SCOPUS database, which allowed us to utilize Boolean operators such as "OR" and "AND". The search results were restricted to (1) journal articles (2) published between 2019 :2023, and (3) written in the English language.

The structured search used to realize the search of the articles was *TITLE-ABS ("Digital Transformation " OR "Digitalization" OR "Digital Twins" OR " Machine Learning " OR "Predictive Maintenance" OR "Internet of Things") AND ("Insurance") AND NOT ("Life Insurance").*

Table 3 Search terms used in the literature search

Variables	Search Terms
Digital Transformation	"Digital Transformation" OR "Digitalization"
Digital Transformation Technologies	"Digital Twins" OR " Machine Learning " OR "Predictive maintenance" OR "IoT".
Business Domain	"Insurance"

Moreover, more keywords were applied to decrease the results. For instance,

TITLE-ABS (" Digital Transformation " OR "Digitalization" OR "Digital Twins" OR " Machine Learning " OR "Predictive maintenance" OR "internet of things") AND ("insurance") AND NOT ("life insurance") AND PUBYEAR > 2018 AND PUBYEAR < 2024 AND PUBYEAR > 2018 AND PUBYEAR < 2024 AND (LIMIT-TO (SUBJAREA , "COMP") OR LIMIT-TO (SUBJAREA , "BUSI") OR LIMIT-TO (SUBJAREA , "DECI")) AND (LIMIT-TO (DOCTYPE , "ar")) AND (LIMIT-TO (LANGUAGE , "English")) AND (LIMIT-TO (EXACTKEYWORD , "Machine Learning") OR LIMIT-TO (EXACTKEYWORD , "Machine-learning") OR LIMIT-TO (EXACTKEYWORD , "Internet Of Things") OR LIMIT-TO (EXACTKEYWORD , "Artificial Intelligence") OR LIMIT-TO (EXACTKEYWORD , "Insurance") OR LIMIT-TO (EXACTKEYWORD , "Digital Transformation") OR LIMIT-TO (EXACTKEYWORD , "Risk Assessment") OR LIMIT-TO (EXACTKEYWORD , "Machine Learning Models") OR LIMIT-TO (EXACTKEYWORD , "Machine Learning Techniques") OR LIMIT-TO (EXACTKEYWORD , "Machine Learning Algorithms") OR LIMIT-TO (EXACTKEYWORD , "Digitalization") OR LIMIT-TO (EXACTKEYWORD , "Risk Management") OR LIMIT-TO (EXACTKEYWORD , "Machine Learning Methods") OR LIMIT-TO (EXACTKEYWORD , "Internet Of Things (IOT)") OR LIMIT-TO (EXACTKEYWORD , "Insurance Companies") OR LIMIT-TO (EXACTKEYWORD , "Predictive Analytics") OR LIMIT-TO (EXACTKEYWORD , "Internet Of Thing (IOT)") OR LIMIT-TO (EXACTKEYWORD , "Machine Learning Approaches") OR LIMIT-TO (EXACTKEYWORD , "Internet Of Things (IoT)") OR LIMIT-TO (EXACTKEYWORD , "Industry 4.0") OR LIMIT-TO (EXACTKEYWORD , "Digital Technologies") OR LIMIT-TO (EXACTKEYWORD , "Insurance Claims") OR LIMIT-TO (EXACTKEYWORD , "Vehicles")) AND (LIMIT-TO (PUBSTAGE , "final")) AND (LIMIT-TO (SRCTYPE , "j")).

B. Database Sources

The primary database chosen for this systematic review was Scopus. This selection was made because Scopus primarily covers scientific conference proceedings and journal articles. Additionally, Scopus was selected as one of the most relevant platforms for accessing scientific information and significant publications across a wide range of academic disciplines.

This database offers advanced search capabilities utilizing logical operators, which align well with the specifications of the proposed systematic review methodology employed in this review. Moreover, Scopus provides useful tools for filtering search results and conducting bibliometric analyses, which are valuable for the comprehensive literature review outlined in this study.

C. Study Selection Criteria

Study selection criteria are specific guidelines or standards used to determine which studies to include or exclude in a research review or analysis that are relevant to the research question or objective. The paper's compliance with the inclusion/exclusion criteria were verified by the reviewers following the conditions out-lined below.

1) Eligibility Criteria

begin, paper contains in its title ("digital transformation" or "machine learning" or "IoT", or "Predictive maintenance" or "digital twins") AND "insurance", This applies to the abstract as well. In situations where the title and abstract were not adequate to determine, the writers reviewed the whole body of the article. To classify if a paper matched these requirements.

2) Inclusion/Exclusion Criteria

In addition to the above, the articles' relevancy was filtered using the following criteria set:

- *Time frame*: Works published from 2019 - 2023 are included.
- *Relevancy*: The existence of keywords (Table 3) in the Title or abstract is crucial for the article's inclusion. On top of that, the papers need to be related to the assessment of DT technology (Digital Twins, Machine learning, IOT, Predictive Maintenance) along the insurance directly.
- *Singularity*: Identical publications identified throughout the database are eliminated.
- *Accessibility*: Strictly peer-reviewed publications that are accessible via the database and are downloadable in full text are included.
- *Language*: Exclusively articles issued in English are listed.
- Subject Area: (computer science, business management accounting, decision science)
- *Document type*: article.
- *Publication stage*: final.
- *Src-type*: journal.

D. Identification

The research process began with a comprehensive search of the Scopus database, yielding an initial set of 9,546 records. This broad search was designed to capture all potentially relevant studies within the scope of the research topic. After retrieving these records, the next step involved applying the inclusion and exclusion criteria to ensure that only relevant studies were considered. This filtering process resulted in the removal of 7,574 records that did not meet the predefined criteria, significantly narrowing the pool to 1,792 records for further analysis.

E. Screening

Following the initial identification phase, the remaining 1,792 records underwent a detailed screening process. During this stage, each record was examined more closely to determine its relevance and suitability for inclusion in the study. Unfortunately, 1,132 records were excluded at this stage, primarily because they were not downloaded and therefore could not be further analyzed. The 660 reports that passed the screening were rigorously evaluated for their alignment with the study's objectives. This assessment led to the exclusion of 580 reports, which were determined to be unrelated to the research focus.

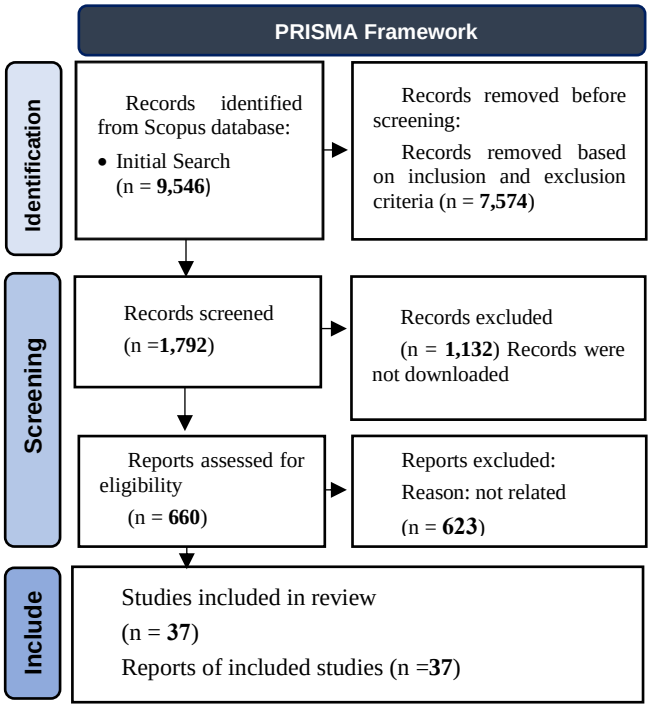

Figure 6 Flowchart detailing the data collection, assessment, and inclusion processes

F. Included

Ultimately, 37 studies were deemed relevant and of sufficient quality to be included in the final review. These studies will form the foundation of the research analysis, providing valuable insights and data to address the research questions.

G. Data synthesis and analysis

To achieve a thorough understanding of our study subject, we employed a dual-method approach that integrates bibliometric analysis with a Content Analysis. This combination allows us to leverage the broad insights gained through biblio-metric analysis alongside the in-depth knowledge provided by the Content Analysis.

1) Bibliometric analysis

Bibliometric analysis was employed as a primary method to explore and answer research questions. Through this approach, the research landscape was systematically assessed, key studies were pinpointed, and trends and patterns within the field were identified. By examining publication and citation data, along with collaboration networks, crucial insights were gained into the evolution and influence of research on digital transformation techniques in insurance.

2) Content Analysis

The extracted data will be systematically synthesized and organized into distinct themes or categories. This process will involve identifying recurring patterns, trends, and insights across the included studies.

VIII. Results

A. The First Phase Bibliometric Analysis:

Bibliometric analysis provides a systematic approach for mapping the intellectual structure of digital transformation in insurance by examining publications, citations, and patterns of collaboration.

Q1**: What is the distribution of studies per year, leading countries, prominent journals, key authors, and most common keywords related to the integration of digital transformation (DT) in the insurance sector**?

To explore the first finding, an analysis of article publication years over the past 5 years was performed. The findings revealed that most publications occurred in more recent years. Notably, in 2023, there were 448 publications on online learning readiness, 348 in 2022, and 226 in 2019. The distribution of the remaining publications was spread across the other years, as depicted in Fig. 7.

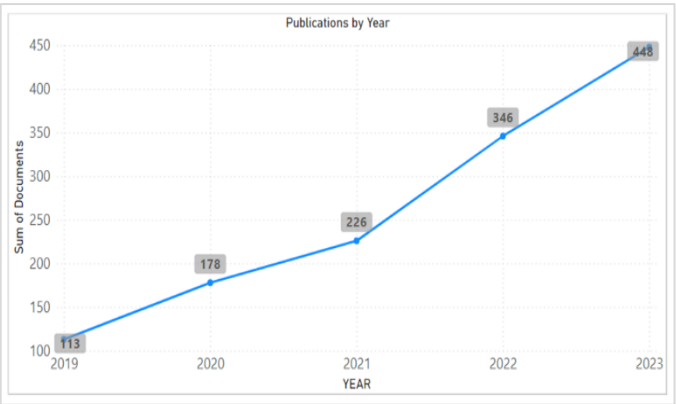

Figure 7 Distribution of publications by years (2019–2023)

A bibliometric analysis of DT in insurance reveals that the United States, China, and India are major contributors to the field, with publication counts of 346, 305, and 220, respectively. The Unit-ed States' leading position in this field can be attributed to its substantial funding, extensive research activities, and a well-established innovation ecosystem in insurance and technology.

The global distribution of research and innovation reflects a diverse and complex approach to integrating DT in insurance, promising a future where educational practices are increasingly personalized, effective, and inclusive, driven by the transformative potential of these technologies as depicted in Fig. 8.

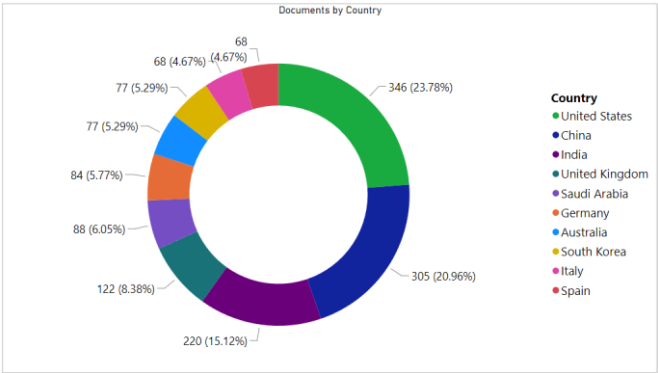

Figure 8 Top 10 Countries Leading in Publications highlights the global impact of digital transformation in the insurance sector.

Fig. 9 presents a network map illustrating the co-authorship relationships between countries in the context of applying digital transformation technologies within the insurance sector. In this map, the size of each node represents the volume of publications from each country, while the lines between circles signify collaborative efforts.

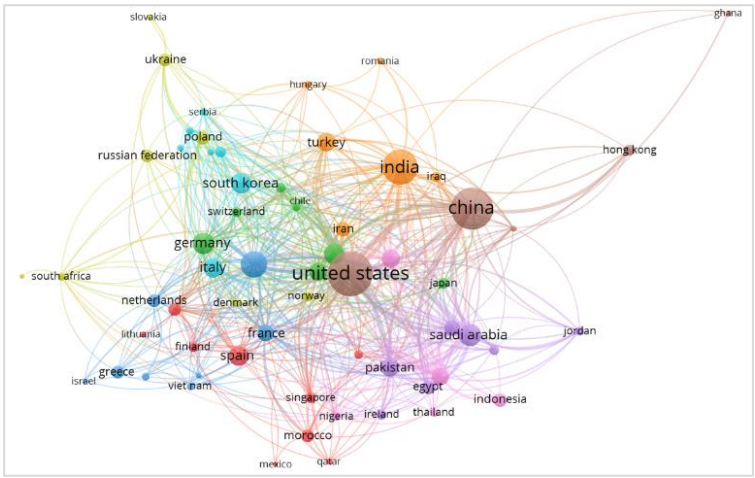

Figure 9 A Network map showing co-authorship relationships between countries

Fig.10 shows the Distribution based on the affiliation, the most affiliated organization were Florida Atlantic University, Vellore institution of technology

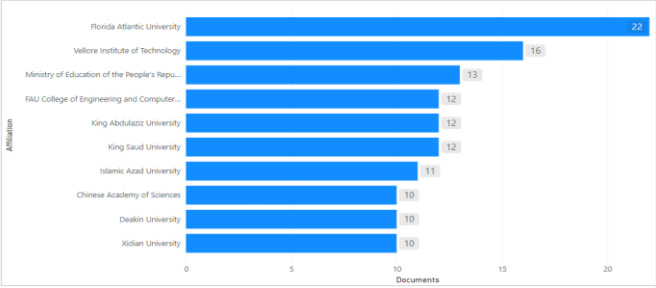

Figure 10 Distribution based on the affiliation

As shown in Table 4, the assessment criteria comprised "Total Publication," "Total Citation", "Journal Cite Score", "Most Cited Article", "Times Cited", and "Publisher" which were based on the bibliometric study of the most cited journals.

According to Table 4, IEEE Access is the most productive journal in the fields of insurance and digital transformation, with 484,743 citations and 49,687 total articles. Sustainability Switzerland comes in second with 381,357 citations and 55,991 publications. Third place goes to Expert Systems with Applications, which has 80,467 citations and 6,464 articles.

Table 5 presents a list of the most prolific authors in this field, showcasing the researchers whose work is shaping the evolving landscape of digital transformation in insurance.

This section examines and highlights the top 10 leading authors who have made significant contributions to digital transformation in insurance. Among them, Taghi M. Khoshgoftaar stands out as a key figure, with an impressive record of 754 publications, an H-Index of 68, and a total of 34,748 citations. He is affiliated with Florida Atlantic University in Boca Raton.

Table 4 Top 10 highly productive journals on DT readiness in the years (2019–2023)

Journal	TP	TC	Cite Score	The Most Cited Article	Times Cited	Publisher
IEEE Access	49,687	484,743	9.8	Metaverse for Healthcare: A Survey on Potential Applications, Challenges and Future Directions	142	IEEE
Sustainability Switzerland	55,991	381,357	6.8	Chatbots in Education and Research: A Critical Examination of Ethical Implications and Solutions	140	MDPI
Expert Systems with Applications	6,464	80,467	13.8	Fusion of U-Net and CNN model for segmentation and classification of skin lesion from dermo copy images	126	Elsevier
Sensors	35,041	256,060	7.3	Survey of Explainable AI Techniques in Healthcare	118	MDPI
Applied Sciences Switzerland	44,530	237,130	5.3	ChatGPT for Education and Research: Opportunities, Threats, and Strategies	188	MDPI
IEEE Internet Of Things Journal	5,781	101,666	17.6	A Survey on the Metaverse: The State-of-the-Art, Technologies, Applications, and Challenges	168	IEEE
Technological Forecasting and Social Change	2,602	55,382	21.3	Digital finance and green growth in China: Appraising inclusive digital finance using web crawler technology and big data	101	Elsevier
International Journal of Advanced Computer Science and Applications	4,997	11,686	2.3	Improved Multiclass Brain Tumor Detection using Convolutional Neural Networks and Magnetic Resonance Imaging	29	Science and Information Organization
Mathematics	15,129	60,616	4.0	A Review of Chaotic Systems Based on Memristive Hopfield Neural Networks	60	MDPI
BMC Medical Informatics and Decision Making	1,353	9,722	7.2	MRI-based brain tumor detection using convolutional deep learning methods and chosen machine learning techniques	98	BMC Medical Informatics and Decision Making

Table 5 The 10 top authors in the field

Authors	TP	TC	H-index	Affiliation	Country
Khoshgoftaar, Taghi Ma	754	34,748	68	Florida Atlantic University The institution, Boca Raton	US
Bauder,Richard A.	29	1,353	19	FAU College of Engineering and Computer Science The institution, Boca Raton	US
Zhu,Quanyan Zhu	393	7,781	43	New York University The institution, New York	US
Leevy, Joffrey L.	36	1,054	14	Florida Atlantic University The institution, Boca Raton,	US
Mercaldo, Francesco	287	5,027	41	Università degli Studi del Molise The institution, Campobasso	Italy
Ming, Ruixing	30	157	6	Zhejiang Gongshang University The institution, Hangzhou,	China
Santone, Antonella	285	3,747	33	Info University degli Studi del Molise The institution, Campobasso	Italy
Srivastava, Gautam	657	16,177	58	Brandon University institution, Brandon,	Canada
Chen, Juntao	58	693	14	Fordham University The institution, New York	US
Delen, Dursun	222	10,143	53	İstinye Universiteit institution, Istanbul,	Turkey

The co-occurrence chart provides a visual representation of the relationships and thematic connections between terms within a research field, helping to identify key topics, trends, and areas of focus. The size of a node in the chart reflects the frequency of the term's appearance: larger nodes indicate terms that are more prevalent. Thicker lines suggest a stronger or more frequent relationship. Different colors often represent different clusters, with each cluster signifying a distinct topic or theme.

A representation of the word co-occurrence network map is shown in Fig. 11. The arrangement depicted in Figure 12 has a total of 652 words. Seven clusters are present. A total of 67,695 pieces makes up the connection strength. also displays phrases that appear in both the titles and abstracts of several publications. An overlay co-occurrence chart provides a dynamic view of how the focus of research has shifted over time, helping researchers to identify emerging trends and potentially underexplored areas within a specific domain. As shown in the figure, the most important technologies being applied in the insurance industry are machine learning (ML), the Internet of Things (IoT), and blockchain.

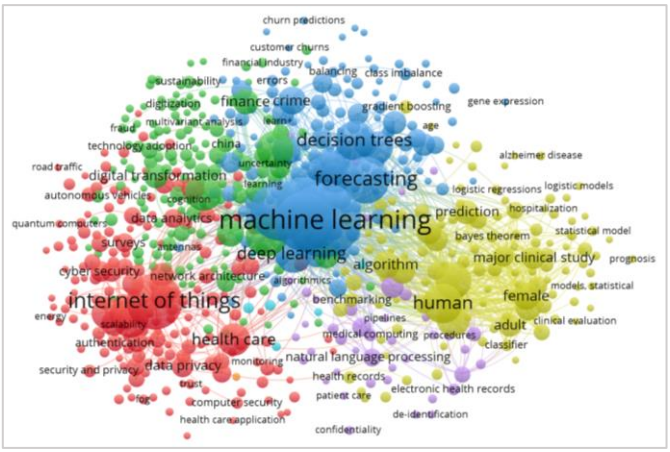

Figure 11 A map based on the relationship of co-occurrence with index keywords

Also evident from Fig. 12 that machine learning and the Internet of Things have been and continue to be among the most important technologies used in the insurance industry from 2020 to 2023. Emerging topics that could be a focus for the future include real-time systems and interaction computer systems. There is a research gap in the application of digital twins and predictive maintenance in the insurance industry.

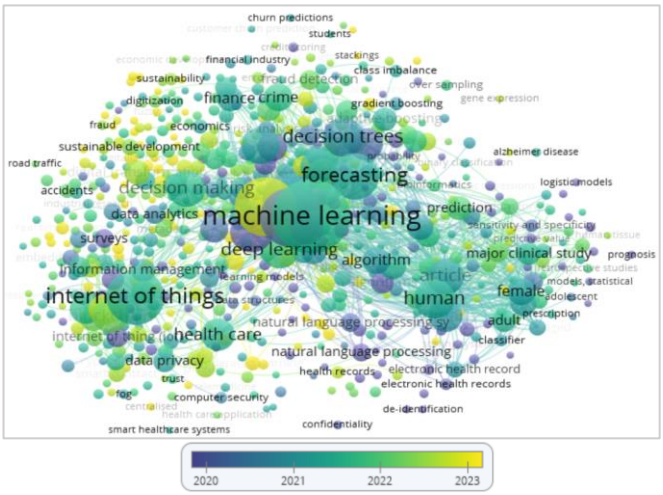

Figure 12 Overlay visualization of 2020–2023

Table 6 summarizes the most frequently used keywords, each appearing at least 5 times, with "machine learning" being the most commonly occurring keyword.

Table 4 Top keywords by the minimum 5 occurrences

Keyword	Occurrences	Total link strength
Machine Learning	487	4807
Learning Systems	310	3258
Machine-Learning	306	3041
Internet Of Things	268	2093
Learning Algorithms	203	2050
Forecasting	189	2312
Artificial Intelligence	149	1527
Human	139	2340
Decision Trees	136	1612
Risk Assessment	122	1276
Decision Making	120	1013
Deep Learning	115	1272

***Q2**: What are the influences of DT technologies in the insurance market?*

This section of the review addresses the second research question by analyzing the impact of digital transformation technologies on the insurance industry, as illustrated in Table 7 in appendix.

The table is organized into two levels: the first level encompasses digital transformation technologies (such as machine learning, the Internet of Things, Digital twins; data analytics.), while the second level details the types of insurance in which these technologies are applied. Machine learning is utilized across various insurance sectors, including cargo insurance, homeowners' insurance, health insurance, auto insurance, general insurance, credit insurance, livestock insurance, asset insurance, and tur-bine insurance. Machine learning aids in enhancing risk assessment, predicting claims, detecting fraud, refining pricing accuracy, analyzing customer sentiments, assessing driving behaviors, personalizing health insurance plans, and understanding disparities in health insurance.

Additionally, digital transformation technologies such as the IoT, and big data allows insurers to collect real-time data on insured assets, enabling them to adjust pricing based on actual risks and user behavior, fostering more personalized and fair pricing strategies. Data analytics enables insurers to identify high-risk individuals and forecast potential claims, allowing for better risk stratification.

***Q3**: What are the key factors influencing the insurance industry's digital transformation?*

This section examines the key variables associated with digital transformation in the insurance industry. Table 8 highlights that "automotive claims" and "fraud detection" are among the most critical variables. Overall, "insurance claims" emerge as a significant area for applying digital transformation techniques, whether in proper-ty insurance or health insurance.

Table 5 Impacts of digital transformation technology insurance industry

References	Objective	Insurance Type	DT technologies	Impact DT on insurance
[A7],[52] [A21],[53] [A64].[54]	Fraud detection	Car Insurance	Machine Learning	– Rise in the quantity of false claims discovered. – Lower the percentage of claims under investigation that are not fraudulent. – Improves risk management by anomaly detection. – Protects privacy by aggregating the real overlapping relationships between vehicle insurance firms on the server. – Reducing the financial losses associated with paying out fraudulent claims.
[A31],[55] [A56],[56] [A63],[9] [A65],[57] [A19],[58] [A50],[59]	Claim prediction	Car Insurance	Machine Learning	– Improve risk assessment; set appropriate premiums; maintain reserves; and precisely predict the incidence of claims. – Improving prediction accuracy for insurance claims. – Increased adaptability and precision in addressing actuarial reserving issues.
[A47],[60] [A54].[61]	Driving Behaviors analysis	Car Insurance	Machine Learning	– Allows for a better understanding of how different behaviors correlate with risk levels in car insurance. – Insurers adopt a more personalized approach to pricing, potentially leading to fairer premiums for drivers who exhibit safer driving behaviors.
[A12],[62]	Vehicle traffic injuries and fatalities prediction	Car Insurance	Machine learning	– Understand collision risks with high accuracy. This leads to better risk assessment, more precise pricing, and improved overall risk management for insurance companies.
[A55],[63] [A61],[64] [A62],[65]	Ratemaking, Predicting Insurance Premium Defaulting	Car Insurance	Machine Learning	– Leads to more precise pricing that reflects current risk levels rather than relying solely on historical averages. – Enables more personalized and fair pricing. – Allowing insurers to detect potential defaults early and take proactive measures. – Enable better risk assessment and management.
[A13],[66]	Aims to predict how often cargo insurance claims occur and the potential severity of those claims	Cargo Insurance	Machine Learning	– Focuses on refining the prediction of insurance claims related to cargo in a specific trade context, aiming to provide more accurate forecasts and better risk management.
[A69],[67] [A74],[68]	The aim of the study is to forecast home insurance claims related to forest fire events, anticipating such incidents to mitigate potential threats to buildings.	House Insurance	Machine Learning	– Save time by making it easier to create insurance plans that function better and more effectively. – Assist in comprehending and reducing the effects of wildfires on residences. – Offer information regarding the effectiveness of preventative actions.
[A9],[69]	Health insurance pricing	Health Insurance	Machine learning	– Make more informed decisions about insurance pricing and coverage
[A14],[70] [A15],[71] [A20],[72] [A25],[73]	Health insurance frauds	Health Insurance	Machine Learning	– Improved fraud detection – Help reduce the manual workload of insurance professionals and improve the speed and consistency of the claim review process. – Benefits include a decrease in processing time for healthcare insurance claims without sacrificing categorization accuracy.

Cont. Table 7 Impacts of digital transformation technology insurance industry

References	Objective	Insurance type	DT Technologies	Impact DT on insurance
[A23],[74]	Aimed to predict polypharmacy using data set of health insurance claims.	Health Insurance	Machine Learning	– Health insurers can more accurately identify individuals at higher risk of requiring multiple medications. This allows for better risk stratification and more personalized health insurance plans.
[A26],[75]	Analyzed client reviews of national health insurance (mobile) applications.	Health Insurance	Machine Learning	– Enhancing Customer Service and Interaction – Developing and Improving Insurance Products and Related Services.
[A38],[76]	Aims to better model and understand disparities in health insurance	Health Insurance	Machine Learning	– Enhances the ability to model and address disparities in health insurance coverage, ultimately leading to better-informed decisions and more equitable health insurance solutions.
[A22],[77]	Conducted customer sentiment analysis and predicted reviews of insurance products using machine learning techniques.	General Insurance	Machine Learning	– Analyzed and predicted customer review messages – Assist businesses in comprehending how clients react to their goods and services.
[A58],[78]	Identified the occurrence of claims in export credit insurance.	Credit Insurance	Machine Learning	– Insurance firms can modify their underwriting policies and raise premium rates in order to counteract the impact of unforeseen claims.
[A59],[79]	Sound deposit insurance pricing	Credit Insurance	Machine Learning	– This helps in better assessing the risks associated with different depositors or financial institutions. – Leading to more effective risk management and pricing strategies
[A29],[80]	Predict buy car insurance or not, detection of health insurance fraud claims	Health Insurance, Car Insurance	Machine Learning	– Marketing strategies based on the identified important features, insurance companies can achieve higher efficiency, better customer engagement, and improved conversion rates, all while optimizing costs. – Enhanced fraud detection
[A33],[81]	Individual insurance costs prediction	Medical Insurance	Machine Learning	– Accurate prediction of medical insurance costs has a transformative impact on the insurance value chain. It enhances underwriting precision, optimizes claims management
[A37],[82]	Detect healthcare insurance fraud	Medical Insurance	Machine Learning	– Enhanced fraud detection
[A1],[83]	Overview of the IoT and connected cars	Car Insurance	IoT, Big Data	– Encouraged insurance businesses to branch out and offer different insurance options, such usage-based insurance, instead of just pure insurance.
[A78],[84]	Proposed a regulatory platform for livestock shed environments to ensure data authenticity and enable real-time monitoring.	Livestock Insurance	IoT, Blockchain	– The automation of the insurer process has decreased costs and increased efficiency, the application of blockchain and the Internet of Things has secured data validity, real-time monitoring, and transparency in the regulatory process.
[A81],[8]	Insurability of cyber physical production systems	Cyber Physical Production	Digital Twins	– Developing new insurance products that can leverage real-time data to monitor the condition and operational status of industrial machines. – Allow insurance companies to adjust their pricing and offers based on the actual risk and performance of the equipment. – Improve predictability of failure risk
[A82],[85]	Enhance the underwriting process for insurance related to complex oil and gas projects	Assets Insurance	Digital Twins	– Reflects a shift towards more data-driven and technologically informed insurance underwriting, which helps in accurately quantifying and mitigating risks in complex oil and gas projects. – Operators involved in exploration and production may now more accurately estimate the likelihood of unfortunate occurrences and reduce the potential for risk.
[A79],[86]	A decade's worth of wind turbine accident news is examined.	Turbine Insurance	Data Analytics	– Risk Assessment and Pricing: By analyzing the types and frequencies of accidents, insurance companies can refine their risk assessment models. This helps in setting more accurate premiums based on the specific risks associated with different types of wind turbines, locations, or operational practices.

Table 8 Illustrates the most important variables regarding DT in insurance

References	Variables of DT	USED TO
[A7],[52] [A21],[53] [A64].[54]	Fraudulent insurance property claims	– The common objective across the research topics listed is to improve fraud detection in insurance using advanced machine learning techniques.
[A31],[55] [A56],[56] [A63],[9] [A65],[57] [A19],[58] [A50],[59]	Motor insurance claim status	– Aims to predict the status of motor insurance claims, such as whether a claim will be approved, rejected, or require further investigation.
[A47],[60] [A54].[61]	Driving behaviors analysis	– Focuses on developing a car insurance model where premiums are determined based on actual driving behavior to determine which driving patterns are most indicative of risk.
[A12],[62]	Vehicle crashes	– Create a ML model that forecasts the likelihood of vehicle crashes, including injuries and fatalities, based on historical crash data.
[A55],[63]	Ratemaking in commercial auto insurance	– Sing machine learning methods to refine the process of setting insurance premiums for commercial auto insurance.
[A61],[64]	Insurance premium defaulting	– The goal is to enhance the prediction of insurance premium defaulting.
[A62],[65]	Auto insurance risk assessment	– Develops a machine learning model specifically for assessing risk in auto insurance.
[A13],[66]	Aims to predict how often cargo insurance claims occur and the potential severity of those claims	– Focuses on refining the prediction of insurance claims related to cargo in a specific trade context, aiming to provide more accurate forecasts and better risk management.
[A69],[67] [A74],[68]	Home insurance claims	– Home insurance claims forecasting, and forest fire occurrences be predicted using algorithmic learning in the event that one of the events poses a hazard to structures.
[A9],[69]	Health insurance pricing	– Examination of training algorithms for the purpose of pricing health insurance.
[A14],[70] [A15],[71] [A20],[72] [A25],[73]	Health insurance frauds	– Develop more explainable machine learning models for fraud detection.
[A23],[74]	Health insurance claims	– Predict the polypharmacy using the health insurance claims data.
[A26],[75]	User reviews of national health insurance	– Examines feedback from customers on the national health insurance application.
[A38],[76]	Disparities in health insurance	– Aims to better model and understand disparities in health insurance using ML.
[A22],[77]	Insurance products' reviews	– Customer sentiment analysis and prediction of insurance products' reviews using machine learning approaches.
[A58],[78]	Occurrence of claims in export credit insurance	– Identifying the occurrence of claims in export credit insurance using machine learning.
[A59],[79]	Sound deposit insurance pricing	– Develop a more accurate and reliable model for pricing deposit insurance by leveraging advanced machine learning techniques.
[A29],[80]	Buy car insurance, detection of health insurance fraud claims	– Using ML and feature selection to provide insurance insights (i.e., identify fraudulent health insurance claims, forecast whether to get auto insurance).
[A33],[81]	Individual insurance costs prediction	– Medical insurance costs prediction using machine learning.
[A37],[82]	Healthcare insurance fraud	– Detect healthcare insurance fraud using machine learning.
[A1],[83]	Connected cars	– Overview of the IoT and connected cars.
[A78],[84]	Livestock insurance	– Using the blockchain system and the internet of things to increase the livestock insurance data's dependability.
[A81],[8]	Insurability of cyber physical production systems	– Enhancing failure risk forecasting by digital twins.
[A82],[85]	Underwriting process for insurance	– Enhance the underwriting process for insurance related to complex oil and gas projects using digital twins.
[A79],[86]	Wind turbine accident news	– Using data analytics, a decade's worth of wind turbine accident news was examined.

Q4: *Which digital transformation technologies has the most used on different segments of the insurance industry?*

This section delves into the key digital transformation technologies that have been adopted within the insurance industry, highlighting their significance and diverse applications. Among these technologies, machine learning (ML) emerges as the most prominent and influential. Fig. 13 vividly illustrates that ML is not only prevalent but also pivotal in transforming various facets of the insurance sector. Its significance is underscored by its broad applicability across multiple insurance types, with particularly notable impacts in automotive and health insurance sectors.

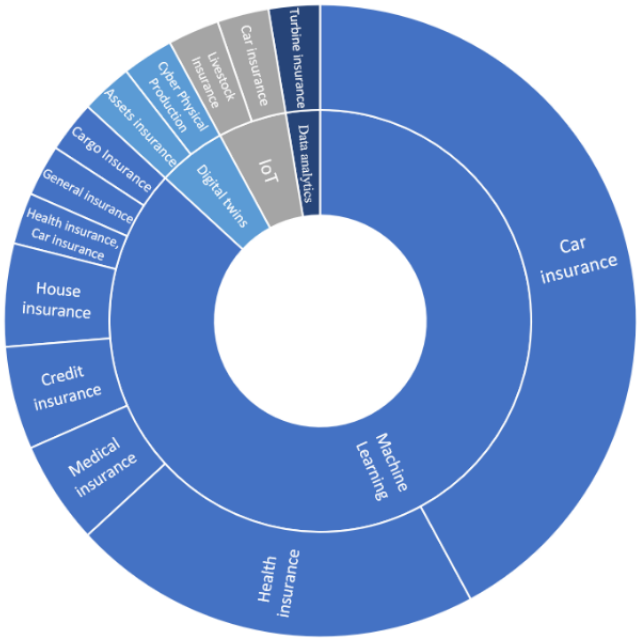

Figure 13 Illustrates the most utilized digital transformation technologies across various sectors of the insurance industry.

Q5**:** *What gaps in the literature exist, and what future research directions can be identified to advance****?***

In our systematic review of the literature on digital transformation (DT) in the insurance industry, several critical gaps have been identified, pointing to areas that require further investigation and development. Firstly, the current body of research reveals a significant under exploration of several key digital transformation technologies, including digital twins, predictive maintenance, data analytics, and the Internet of Things (IoT). While some studies have touched on these technologies individually, there is a lack of comprehensive research into their collective impact and implementation across the insurance sector. This gap indicates a need for more focused studies that explore how these technologies can be integrated and leveraged to enhance various aspects of insurance operations and risk management.

Secondly, specific types of insurance, such as equipment insurance, present unique challenges that have not been adequately addressed by existing digital transformation initiatives. Equipment insurance, which covers machinery and technical assets, faces difficulties in adopting and applying digital transformation technologies effectively. These challenges include the complexity of assessing risks related to machinery and the difficulty of predicting maintenance needs using traditional methods. As a result, there is a noticeable deficit in the application of advanced technologies like machine learning, IoT, and predictive maintenance in this area.

To advance the field, future research must address these gaps by examining how digital transformation technologies can be utilized to bridge these shortcomings within the insurance value chain. By exploring the potential of digital twins, IoT, and predictive maintenance, researchers can uncover new ways to enhance risk management and operational efficiency. These technologies hold promise for creating innovative business models that go beyond traditional risk financing. For example, digital twins can provide real-time simulations of equipment performance, IoT can offer continuous monitoring and data collection, and predictive maintenance can anticipate and address issues before they lead to significant claims. This shift from a reactive to a proactive approach in risk management could significantly transform the role of insurance companies, allowing them to move from merely financing risks to actively preventing them and adding value through advanced risk management solutions.

IX. Conclusion

This review underscores the critical challenge faced by insurance companies in navigating the digital transformation landscape amidst rapid technological advancements and competitive pressures. Despite some progress, many insurers have been slow to adapt to new technologies, which has resulted in missed opportunities for enhancing risk management and operational efficiency.

The paper reveals significant gaps in the current literature, particularly regarding the application of digital transformation technologies such as digital twins, predictive maintenance, and advanced data analytics in the insurance sector. The review highlights the widespread use of machine learning across various insurance types, emphasizing its role in improving risk assessment, claim prediction, and fraud detection. Technologies like the Internet of Things (IoT) and big data are also instrumental in enabling real-time data collection and personalized pricing strategies.

However, the study identifies a critical gap in the application of these technologies to specific insurance sectors, such as equipment insurance, where traditional methods fall short. To address these challenges, future research should focus on integrating digital transformation technologies into these underexplored areas, aiming to shift the insurance industry from a traditional risk financing model to a more proactive risk management approach.

In conclusion, this review contributes valuable insights into the role of digital transformation in the insurance industry and offers a forward-looking perspective on how emerging technologies can reshape the sector. By highlighting existing gaps and suggesting future research directions, the paper provides a foundation for advancing both theoretical and practical understanding of digital transformation in insurance.

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